8.3-inch 1058-ppi Crystal Indium Oxide Display with Side-by-side Method Using Fine Metal Mask and Partition Wall Dividing OLED between Sub-pixels

Shingo Eguchi, Tomoya Aoyama*, Sho Kato*, Takayuki Oide*, Seiko Inoue*, Sachiko Yamagata*, Daisuke Kubota*, Koji Kusunoki*, Satoru Idojiri**, Yasutaka Nakazawa**, Rai Sato** and Shunpei Yamazaki**

***Semiconductor Energy Laboratory Co., Ltd., Kanagawa, Japan**

****Semiconductor Energy Laboratory Co., Ltd., Tochigi, Japan**

Abstract

We have developed a technology enabling high-resolution organic light-emitting diode displays exceeding 1000 ppi with crystal indium oxide backplane, which does not require photolithographic patterning. Introducing a partition wall between sub-pixels and employing our novel OCTILE® pixel arrangement allow application of a fine metal mask process to high-resolution displays.

Author Keywords

OLED; SFM2; OCTILE; pixel arrangement; Natural 3D; crystal IO

1. Introduction

With the recent development of organic light-emitting diode (OLED) technologies, mass-produced OLED displays of various specifications have been appearing on the market. OLED displays have increasingly become popular in a wide variety of fields, including portable information terminals such as smartphones and tablet PCs, small displays for virtual reality and augmented reality devices, and large displays for TVs.

Displays for smartphones and tablet PCs, which generally have screens with a diagonal size of approximately 10 inches and a resolution of approximately 500 ppi, have mostly been replaced by OLED displays. This is because the displays for these applications are direct-view displays, which are manufactured with high productivity through OLED patterning using a fine metal mask (FMM). However, it is difficult for an FMM to achieve higher-resolution displays required for further enhancement of the image quality of smartphones and tablet PCs.

As a process for solving this problem, we have developed a metal mask-less lithography (MML) technology [1-3]. With this technology, an OLED is uniformly formed on the entire display screen area, and then the OLED for a pixel of a target color is left and the other unnecessary OLEDs are removed by photolithographic patterning and etching. This process is performed for three colors of red (R), green (G), and blue (B). In addition, we have been developing displays with varying resolutions based on our crystal indium oxide (IO) field-effect transistor (FET) technology [4-6]. Our technology is applicable to displays of various specifications without limitation by the substrate size, the substrate material, or the display resolution, as long as photolithography can be employed. Utilizing this technology, we have fabricated a 1058-ppi display using a glass substrate and a display exceeding 5000 ppi using a silicon substrate [7-8]. However, since OLEDs for RGB must be formed, the MML technology requires three times repetition of vapor deposition and patterning, and also needs photolithographic patterning equipment in addition to the existing vapor deposition equipment.

In consideration of the above, it is important to perform ultrahigh-resolution OLED patterning using an FMM by effectively utilizing the vapor deposition equipment that has been widely employed for the mass production of high-resolution OLED displays. To achieve this, we have been developing a super fine metal mask (SFM2) technology that enables fabrication of high-resolution displays exceeding 1000 ppi without requiring additional equipment.

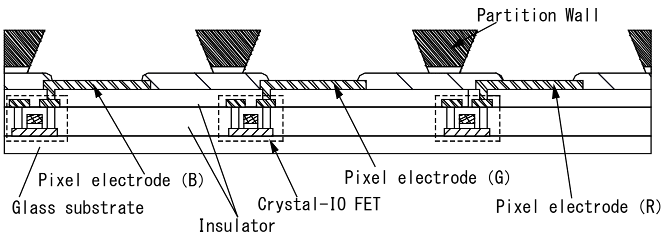
2. Overview of SFM2 Technology

Figure 1 shows the process for the SFM2 technology. The process features the use of a backplane substrate with partition walls having inverted triangular cross-sectional shapes on pixel definition layers between sub-pixels, as shown in Figure 1(a). When OLED organic layers are formed by vapor deposition using an FMM on the substrate in the subsequent process as in Figure 1(b), the OLED organic layers are divided by the partition walls for the respective sub-pixels. After that, cathodes are formed as shown in Figure 1(c), so that OLED devices are completed.

The reason why the partition walls have the inverted triangular cross-sectional shapes is to divide the organic films formed on the partition walls and to separately pattern the organic films on the individual pixel electrodes. In OCTILE® arrangement described later, adjacent sub-pixels of the same color are collectively formed with one opening pattern of an FMM, and an organic film for this part must be divided by partition walls. As shown in Figure 2, one partition wall has a stacked-layer structure of an insulating layer and a conductive layer. Among OLED layers, an organic film layer such as a light-emitting layer is divided by a partition wall for each pixel. Even when the evaporated film is deposited in the shadow of the partition wall and thus part of the organic film comes in contact with the bottom of the partition wall, the insulating layer in the lower portion of the partition wall inhibits the electrical contact between the organic film and the partition wall. By contrast, a layer near the cathode is in electrical contact with the conductive layer of the partition wall due to its coverage. All the pixels in the screen area are connected to a cathode wiring through the cathodes and the partition walls to configure an OLED circuit as a whole. This structure suppresses lateral leakage between the adjacent pixels, thereby preventing a voltage drop due to the resistance of the thin cathodes. Note that the partition wall does not necessarily have the inverted triangular shape as long as the above-mentioned function can be fulfilled. For example, a cross-

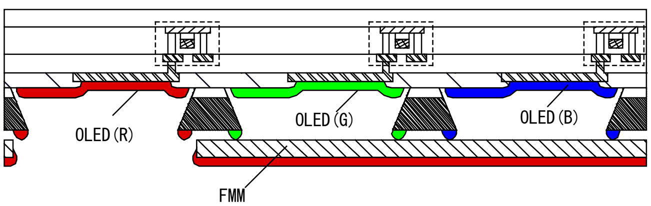
ISSN 0097-996X/26/5701-1024-$1.00 © 2026

sectional shape shown in Figure 3 provides the equivalent effect. Figures 4(a) and 4(b) show inverted tapered partitions of Types A and B, respectively.

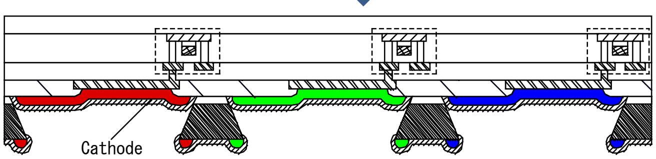


(a) Backplane with partition wall

Inverted
Using FMM
For three colors of RGB



(b) FMM OLED patterning (for three colors)



(c) Cathode deposition

Figure 1. Process for SFM2 technology.

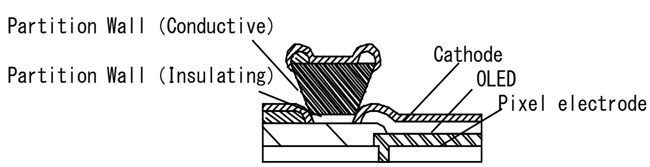


Figure 2. Deposition state around partition wall (type A)

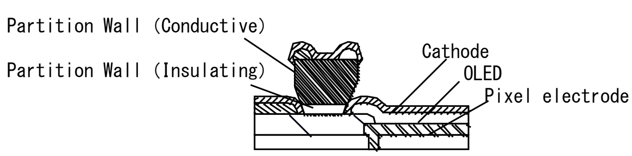
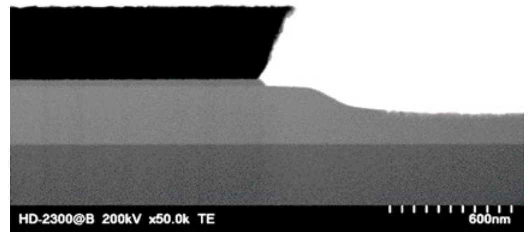
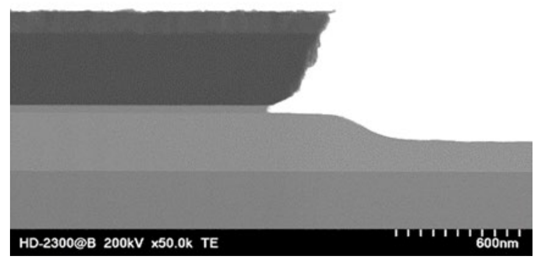


Figure 3. Variation of partition wal (type B)



(a) Type A



(b) Type B

Figure 4. Cross-sectional shapes of partition walls.

3. New Pixel Arrangement

The FMM pixel patterning is often employed for displays having resolutions of 500 ppi or lower. In recent years, development aiming for application of the FMM pixel patterning to displays exceeding 500 ppi has been progressing. However, it is necessary to leave the alignment margin for vapor deposition using the FMM and to avoid vapor deposition shadowing, which creates the need for taking measures such as a reduction in the sub-pixel aperture ratio for a wide space between pixels. This measure leads to high current stress against OLED devices necessary for exhibiting predetermined luminance, shortening the emission lifetimes of the OLED devices.

Aiming at the application of the FMM patterning to OLED displays with a resolution of approximately 1000 ppi, we employ sub-pixel arrangement shown in Figure 5 in which sub-pixels of the same color for four pixels are gathered in one place to be collectively patterned with one opening pattern of an FMM as indicated by dotted lines in Figure 5. The repeating unit of 2 × 2 pixels includes two R sub-pixels, four G sub-pixels, and two B sub-pixels. We call this sub-pixel arrangement "OCTILE®". The space between sub-pixels of different colors is 10 μm or more in consideration of FMM patterning margin. Meanwhile, since sub-pixels of the same color are collectively patterned with one opening pattern of the FMM, which eliminates the need for considering the FMM patterning margin, the space between the sub-pixels of the same color is several micrometers though there is the partition wall in the space. Such pixel arrangement allows fabrication of a 1000-ppi display using an FMM with an aperture pitch corresponding to 500 ppi. The OCTILE® arrangement enlarges the area of each sub-pixel, and an aperture ratio increase reduces the current stress on OLEDs. Note that the above-mentioned partition wall exists not only between the sub-pixels of different colors but also between the sub-pixels of the same color, and contributes to a reduction in lateral leakage between the adjacent pixels.

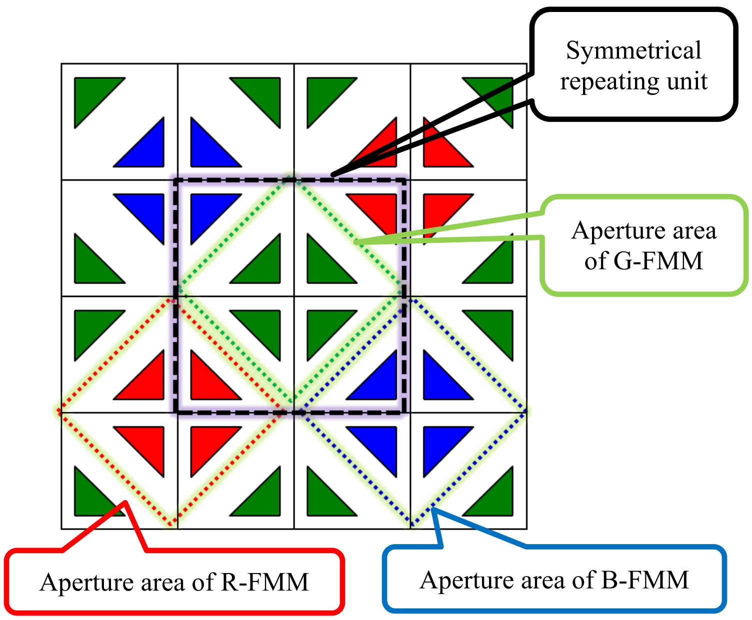


Figure 5 OCTILE® pixel arrangement.

4. Display Specifications and Natural 3D Comparison

Figure 6 shows a displayed image of the prototype. Table 1 lists the specifications of the display.

Figure 6. Displayed image of 1058-ppi 8K OLED display with OCTILE® arrangement prototyped this time.

Table 1. Display specifications.

Items	Specifications
Display structure	OLED + Crystal IO (glass substrate)
Display area size	8.3 inches diagonal 184.32 mm (H) × 103.68 mm (V)
Pixel count	7680 × 4320 (G) (7680 × 4320)/2 (R, B)
Pixel size	24 µm × 24 µm
Resolution	1058 ppi
Pixel arrangement	OCTILE®
Aperture ratio	15.9%
Frame frequency	60 Hz
Driver circuit	Source: IC bonding Scan: integrated

The sub-pixel arrangement is significantly different between the OCTILE® arrangement and the stripe arrangement, which is generally employed for displays; thus, the display with the OCTILE® arrangement may show an image different from that shown by the display with the stripe arrangement.

Our previous 1058-ppi 8K OLED display with the stripe arrangement is a direct-view display showing a two-dimensional video; however, the video sometimes offers a stereoscopic effect.

We call this phenomenon "Natural 3D (N3D)" [9]. The N3D is probably derived from the principle where two-dimensional optical information from a display enters eyes and is then converted into three-dimensional information in the visual cortex of the occipital lobe of the brain to be recognized stereoscopically. That is, the N3D is induced by brain illusions.

There are various factors causing brain illusions [10]. The N3D features that when a person watches one display, the same two-dimensional video enters both eyes. Thus, physiological factors such as binocular disparity and binocular convergence, among keys to obtaining stereoscopic effects, cannot be used. The N3D utilizes psychological and experimental factors such as perspective and motion parallax. When a person naturally watches a direct-view video, the N3D offers stereoscopic effects without requiring special mechanisms. Hence, the person hardly gets motion sickness or eye strain even when watching the video for a long time.

We examined the N3D effect on an 8.3-inch 1058 ppi OLED display fabricated with the developed OCTILE® arrangement. Various volunteers watched a video on the fabricated display under luminance of 250 cd/m^2 and viewing distance of 45 cm, and then answered the following question: “Did you perceive the video stereoscopically?” Figure 7 shows the survey results.

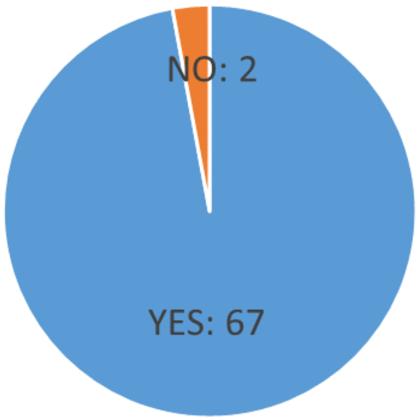


Figure 7. Answers to question: "Do you feel N3D stereoscopic effect of prototyped display with OCTILE® arrangement?"

According to the results in Figure 7, the 1058-ppi display with the OCTILE® arrangement shows the video offering the stereoscopic effect, which reveals that the N3D is exhibited without any problems.

5. Conclusion

We have developed the SFM2 technology aiming at an OLED display exceeding 1000 ppi using an FMM. The combination of the partition walls and the OCTILE® pixel arrangement achieves fabrication of a display having a pixel density difficult to realize by the FMM patterning, with the use of the existing vapor deposition equipment. Employing this technology, we have prototyped the display exceeding 1000 ppi, which exhibits the N3D. The SFM2 technology is nearly completed. Our future works are to reduce the cost and to enhance the performance of displays by utilizing the SFM2 technology providing high mass productivity.

6. Acknowledgement

The authors would like to thank Canon Tokki Corporation for their helpful advice on improving the evaporation equipment and for their tremendous support in this work.

References

[1] S. Eguchi et al., "An 8.3-in. 1,058ppi OLED Display with Side-by-Side Pixel Structure Fully Fabricated by Photolithography," SID Symp. Dig., 54, pp.209-212 (2023)

[2] Y. Yamane et al., "3207-ppi, 1.50-in. OLED Microdisplay with All Pixels Formed Through RGB Side-by-Side Patterning by Photolithography" SID Symp., Dig., 54, pp. 1334-1337 (2023)

[3] N. Sugisawa et al., "High Performance OLED with Microlens Array, Metal Mask-Less Lithography, and RGB Side-by-Side Patterning" SID Symp., Dig., 55, pp. 844-847 (2024)

[4] Y. Shima et al., "Normally-off top-gate self-aligned field-effect transistor using crystal InOx with field-effect mobility of around 100 cm^2/vs." J. Soc. Inf. Disp., 33, pp. 411-424 (2025)

[5] Y. Nakazawa et al., "Trifold OLED Display Fabricated through Low-Temperature Process Using Short Channel Top-Gate Self-Aligned Field-Effect Transistor with Crystal IO" SID Symp., Dig., 56, pp. 1044-1047 (2025)

[6] S. Yamazaki et al., "High-performance single-crystalline In2O3 field effect transistor toward three-dimensional large-scale integration circuits" Commun. Mater. 5(1):184 (2024)

[7] M.Kozuma et al., "1.5-inch, 3207-ppi Side-by-Side OLED Display Capable of 32-Division Driving with OSLSI/Si-LSI Structure Fabricated by Photolithography" SID Symp., Dig., 53, pp.384-387 (2022)

[8] Y. Tamatsukuri et al., "5009-ppi, 10000-cd/m2, OLED/OS/Si Structure Display with Built-in CPU and Display Driver" J. Soc. Inf. Disp., 33, pp. 314-323 (2025)

[9] Y. Yanagisawa et al., "Curved OLED Display to Effectively Enhance Natural3D" SID Symp., Dig., 46, pp. 1263-1266 (2015)

[10] Y. Iwadate et al., NHK Science & Technology Research Laboratories R&D No.151 (2015)